# 9.2 Populations, Samples, and Hypotheses

Name _____ Period ____ Date _____

Algebra II 9.2 · Student sheet

**I can…**

- tell the difference between a population and a sample, and between a parameter and a statistic.
- use a simulation to find proportions.
- use a simulation to decide whether to reject a hypothesis.

**Vocabulary** · fill in the missing word

**Population** — the entire group you want _____ about

**Sample** — a part of the _____

**Parameter** — a number that describes a _____

**Statistic** — a number that _____ a sample

**Hypothesis** — a claim about a _____ of a population

**Simulation** — a model of a _____ situation, often using random numbers or coins

## Part 1 · Is the coin fair?

A friend got 9 heads in 10 flips and says the coin is lucky. Flip a fair coin 10 times. Heads: _____

| Heads | 0–3 | 4 | 5 | 6 | 7 | 8 | 9–10 |
|---|---|---|---|---|---|---|---|
| Class count | | | | | | | |

How often did a fair coin give 9 or 10 heads? _____ Is 9 heads unusual for a fair coin? _____

## Part 2 · Notes and examples (fill in each note, then work the example)

**Population:** _____. **Sample:** _____.

### Example 1

A school surveys 100 of its 1,200 students about the lunch menu. Identify the population and the sample.

**Parameter or statistic:** A **parameter** describes a _____; a **statistic** describes a _____.

### Example 2

Tell whether each number is a parameter or a statistic.

a. The mean age of all 30 members of a club is 34.

b. In a survey of 1,000 voters, 52% plan to vote yes.

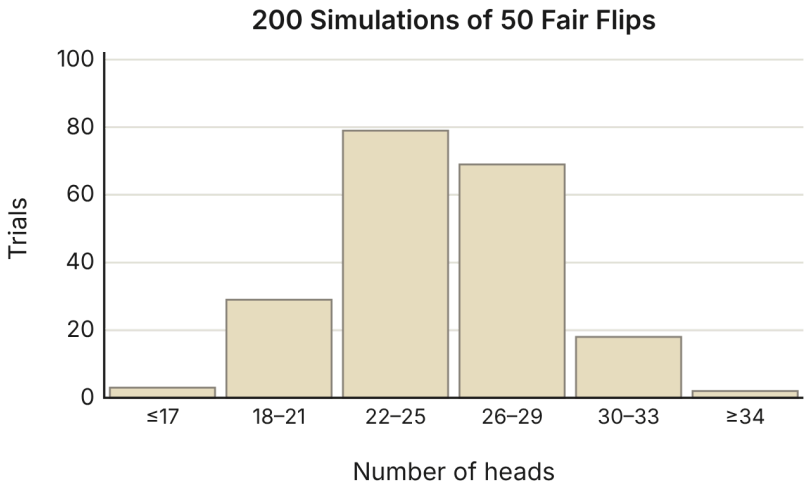
### Example 3

A spinner is spun 50 times and lands on red 18 times. What proportion of the spins landed on red?

**Testing a hypothesis:** Reject a hypothesis when the result is very _____ if the hypothesis is true.

### Example 4

You claim a coin is fair. In 50 flips, you get 34 heads. A simulation of 200 trials of 50 fair coin flips gave 34 or more heads in only 2 trials. What should you conclude?




## Part 3 · Partner practice

### Partner 1

Tell whether each data set is a population or a sample.

a. the ages of every person counted in a town's census

b. the heights of 30 randomly chosen 10th graders

**Partner 2**

Tell whether each number is a parameter or a statistic.

a. A poll of 500 shoppers finds that 40% prefer store A.

b. The 25 players on a team average 6 feet tall.